B.C.A. 1st Sem. Examination, 2025 (Session : 2024-2027)
Fundamentals of Computer (Paper : 1 BCA1)
Paper Code : 111601

Time : Three Hours **[Full Marks : 100 Pass Marks : 40]**

Note : Candidates are required to give their answers in their own words as far as practicable. The questions are of equal value. Answer any five questions.

1. What is Disk Operating System? How does it manage File? Discuss with file related commands.

2. Differentiate between any two of the following :

(i) Single User Vs. Multi user system.

(ii) Decimal Vs. Binary number system

(iii) 9's Vs. 10's complement

(iv) Static RAM Vs. Dynamic RAM.

3. With pictorial representation discuss Von Neumann architecture of Computer and its working behaviour.

4. Discuss Digital and Analog computer with their features and examples. Also write some major differences.

5. Explain different binary arithmetic operations using 1's & 2's complements with example.

6. Convert the following numbers to decimal:

(i) $(ACD)_{16}$

(ii) $(1111.11)_{2}$

(iii) $(756)_{8}$

(iv) $(CD)_{16}$

(v) $(BCA)_{16}$

7. Discuss the architecture of Hard Disk Drive? How does it work? Also discuss seek time, latency time and access time.

B.C.A. 1st Sem. Examination, 2025 (Session : 2024-2027)
PC Packages (Paper : 1BCA2)
Paper Code : 111602

Time : Three Hours **[Full Marks : 100 Pass Marks : 40]**

Note : Candidates are required to give their answers in their own words as far as practicable. The questions are of equal value. Answer any five questions.

1. What is macro? How is a macro created?

2. What is function and formula in MS Excel? Discuss some functions and formulas. How is a nested function used in MS-Excel? Explain with example.

3. Describe 10 different GUI components.

4. What is disk defragmentation? Also discuss formatting of hard disk.

5. Explain the process of the following features of Word Processing MS-Word software:

(a) Mail Merge features
(b) Pivot table
(c) Goal seek
(d) Page setup

6. Explain the role of operating system in memory management.

7. What are the various types of charts available in MS-Excel? Explain their uses.

8. Write steps for inserting watermark, header and footer, hyperlink.

9. Write the steps for:

(i) Removing duplicate data
(ii) Filter data
(iii) Freezing pane
(iv) Page orientation

B.C.A. 1st Sem. Examination, 2025 (Session : 2024-2027)
Programming Logic & Design Techniques (Paper : 1BCA3)
Paper Code : 111603

Time : Three Hours **[Full Marks : 100 Pass Marks : 40]**

Note : Candidates are required to give their answers in their own words as far as practicable. The questions are of equal value. Answer any five questions.

1. What is time complexity? Discuss Worst case, Average case, Best case and different notations.

2. How features of Algorithm help us for solving a problem in the best way?

3. What is space complexity? How is it measured?

4. What is inheritance? Discuss features and types of inheritance.

5. Explain procedural and recursive programming.

6. Explain nesting of IF-ELSE statement using an algorithm.

7. Design a flowchart for representing decision making statements. Discuss different symbols used in flowchart.

8. What is pseudo code? Write a pseudo code for printing table of 5 (five).

9. Write short notes on any two of the following:
- (a) Variable
- (b) Constant
- (c) Data type
- (d) Encapsulation

10. What is iteration and selection control structure?

B.C.A. 1st Sem. Examination, 2025 (Session : 2024-2027)
Programming in C (Paper : 1BCA 4)
Paper Code : 111604

Time : 3:00 Hours **[Full Marks : 100 Pass Marks : 40]**

Note : Candidates are required to give their answers in their own words as far as practicable. The questions are of equal value. Answer any five questions.

1. Define Array. Write a program in C to find the sum of odd and even numbers in a given array.

2. What is an operator? Discuss various types of operators, also discuss precedence and associativity of operators.

3. Write a program in C to calculate the length of a string without using library function.

4. Explain storage class modifiers with all its features. Also give suitable examples.

5. What is format specifier? Write the uses of different format specifiers with example.

6. Write a program in C to check whether inputted number is perfect or not.

7. What is selection control structure? Discuss in detail with example.

8. Differentiate between the following:
(i) while and do-while loop
(ii) break and continue
(iii) automatic and static variables
(iv) getch() and getche()

9. What is an operator? Discuss relational and logical operator with example. Also discuss unary and binary operator.

10. Write short notes on any two of the following with example:
(i) Continue
(ii) Identifier
(iii) Escape sequence character

B.C.A. 1st Sem. Examination, 2025 (Session : 2024-2027)
Communicative English (Paper : 1BCA 5)
Paper Code : 111605

Time : 3:00 Hours **[Full Marks : 100 Pass Marks : 40]**

Note : Candidates are required to give their answers in their own words as far as practicable. The questions are of equal value. Answer any five questions.

1. Prepare your Curriculum Vitae (CV) for the role of Sales Manager in a reputed company citing all your qualifications, experiences, work history, and salary expectations.

2. Give a project plan for the establishment of a cultural club in your college.

3. Rewrite the following sentences into passive voice:

(a) Bhim ate eight chapaties at lunch.
(b) He brought fruits from Gola.
(c) Priyam presented the report yesterday.
(d) I have done the work successfully.

4. Write an invitation to the Union Minister for IT to attend an international seminar organized by your department.

5. What is presentation? Discuss skills required for effective communication.

6. Write an article on any one of the following in approx 200 words:

(a) Road safety
(b) Digital Arrest

7. How is group discussion different from an interview? Which mode do you prefer? Give reasons for your choice.

8. What is Communication? Delineate various kinds of communication. Discuss barriers to communication.

9. Group Discussion generates opportunity for improving speaking skills. Discuss speaking skills.

10. Answer the following in 25 to 50 words:

(a) Notice for staff meeting
(b) What is summary?
(c) Why is pitch of voice important in communication?
(d) Overcoming nervousness